Complaint Root-Cause Engine

ExaThon 2026 Build Plan – Track 01: Money, Risk & Regulation

The Official Record

400

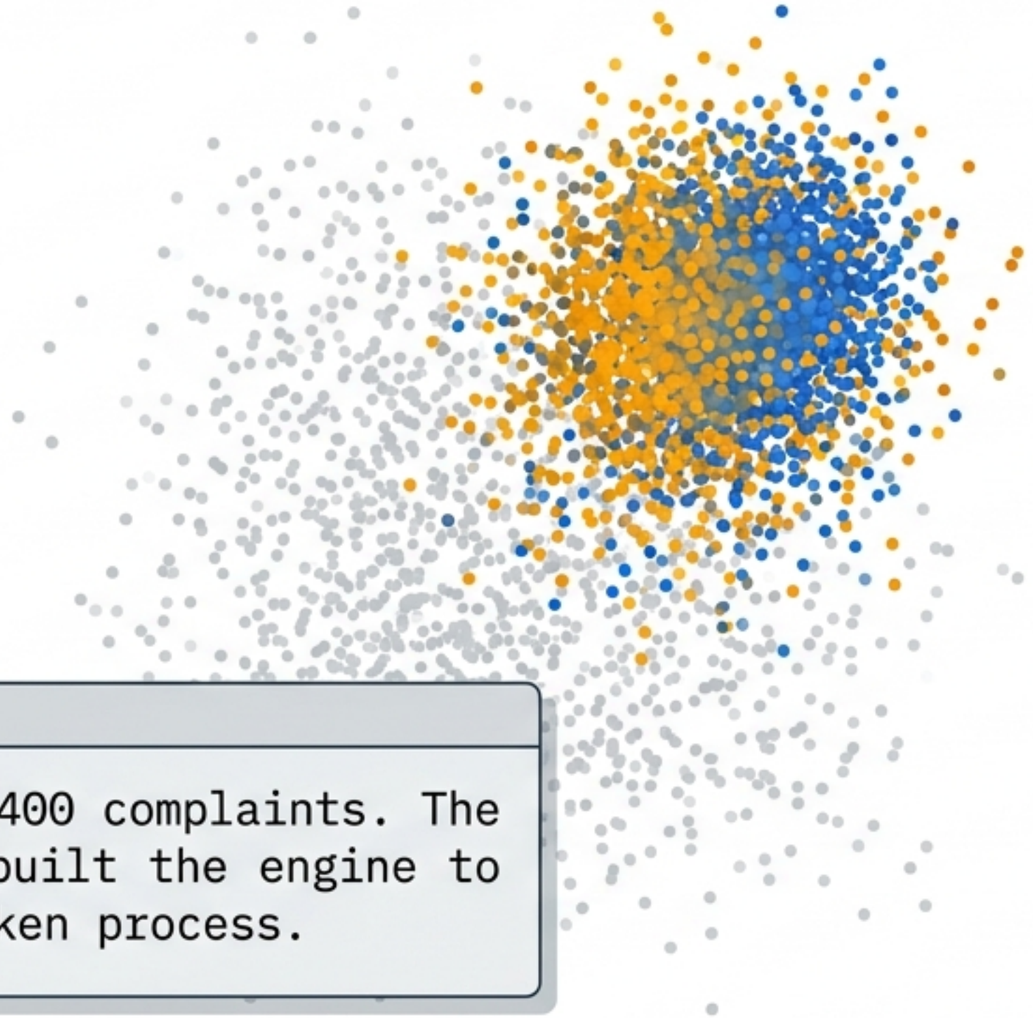


Complaints Last Month (Agent Tagged)

The Ground Truth

1,100

Actual Complaints Detected in Free Text



Terminal

```
Agent tags say the bank had 400 complaints. The
text says it had 1,100. We built the engine to
find the 600 hiding in a broken process.
```

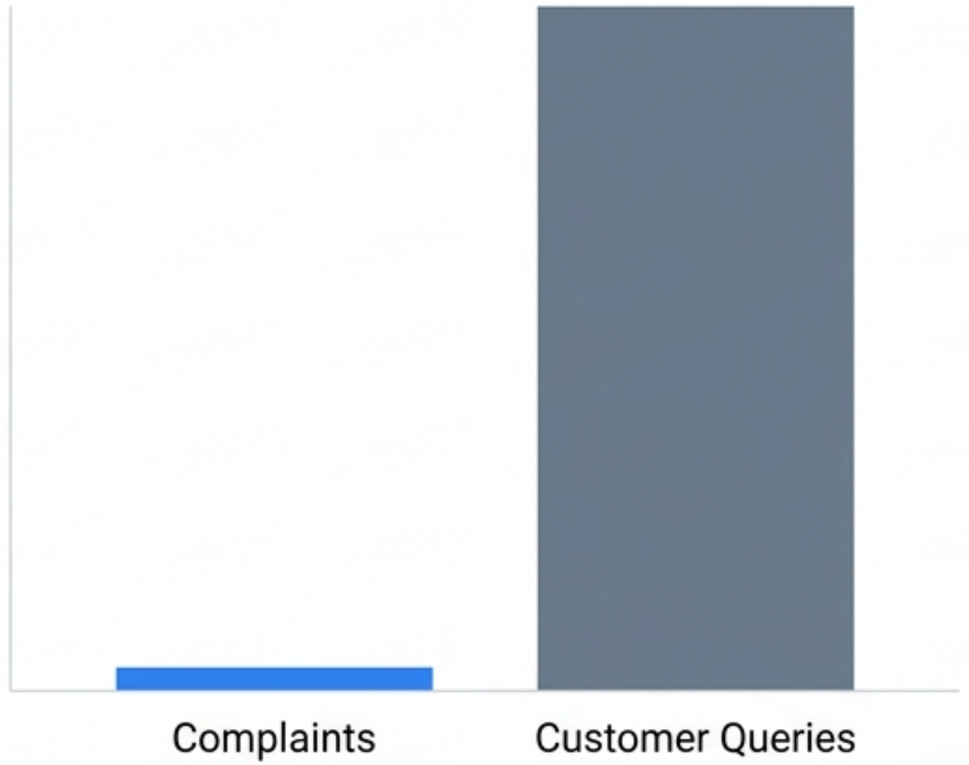
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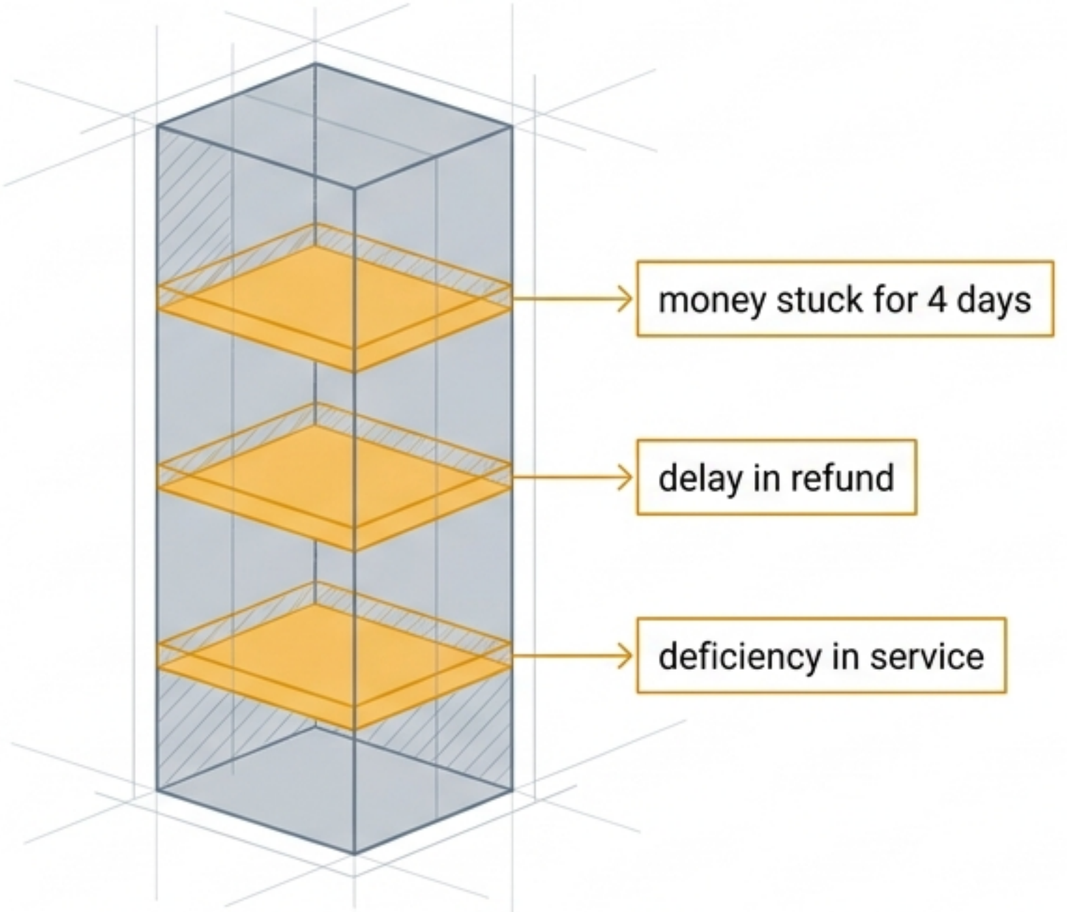
The Blind Spot of Manual Classification

The internal ombudsman receives thousands of tickets across branches, call centres, and the app.

She is blind to systemic process failures because categories are inconsistent and the true reasons are buried in free text.

Standard CRM dashboard





The Result: Banks answer the same complaint 500 times instead of fixing the root cause once.

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The Regulatory Squeeze on Grievance Redressal



13.34 Lakh

Complaints under RBI's scheme in FY 24-25, representing a 13.55% volume spike.



Governor's Warning

RBI Governor explicitly flagged banks misclassifying real complaints as customer queries.



Root-Cause Mandate

Deputy Governor demands banks conduct deep root-cause analysis, not just surface tracking.



The Compliance Hook

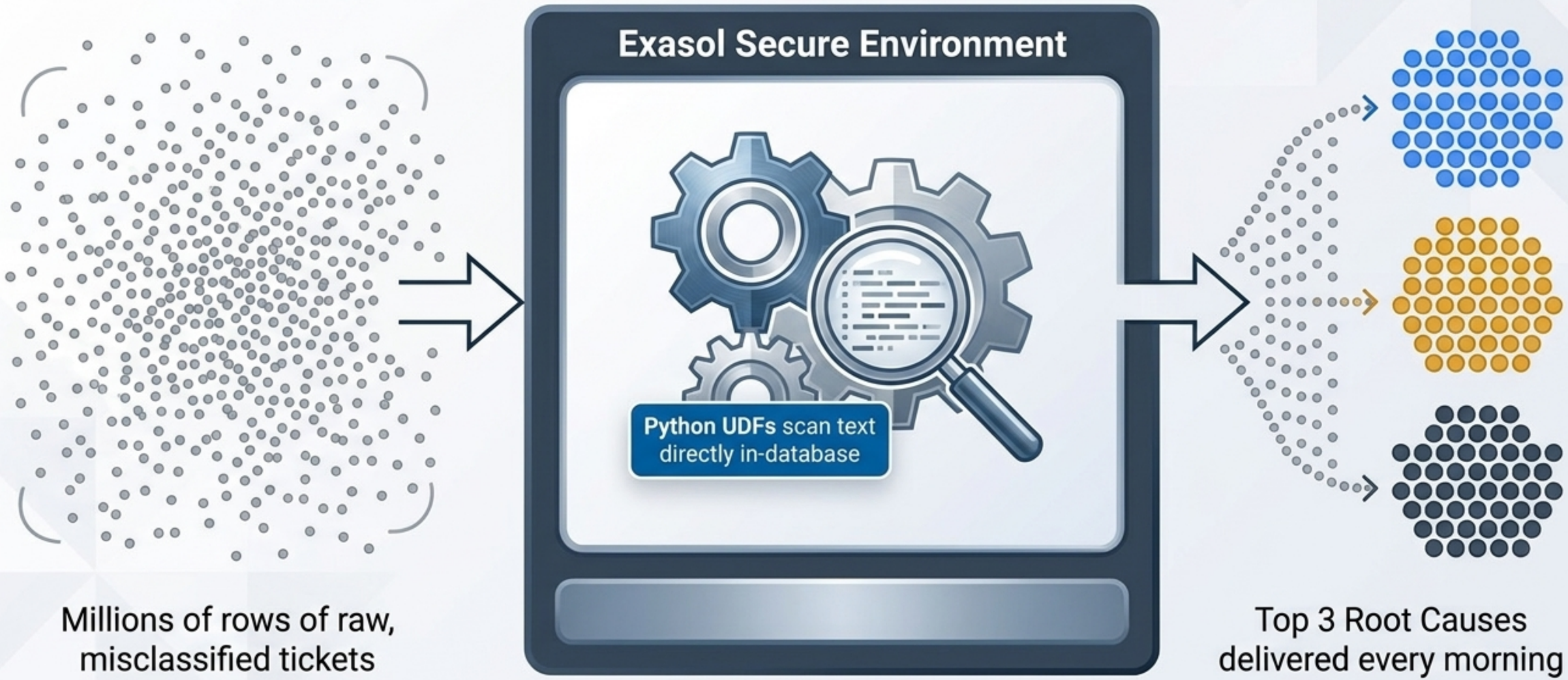
Mandatory disclosure of top 5 complaint grounds using the RBI master list. Offenders face review.

Banks miss systemic problems until the RBI sees them.
This engine is the direct antidote to escalating mandates.

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Sovereign Analytics: X-Raying the Database

Sovereign Analytics: The ticket text never leaves the bank's database.



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The Ombudsman's Diagnostic Dashboard

Turning fragmented text into fixable process failures and direct compliance outputs.

Root-Cause Board



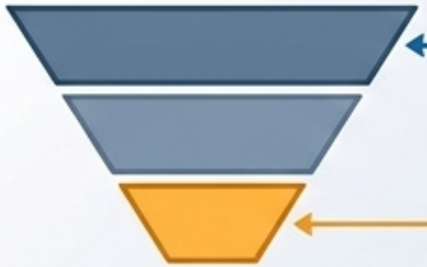
The three causes driving most complaints at a glance.

Misclassification Watchlist



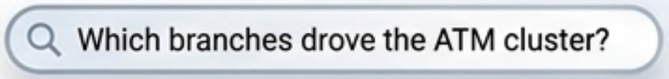
Isolates the exact misclassification issue the RBI Governor flagged.

Escalation Gap



Catches improperly rejected grievances that never reached the ombudsman.

Ask the Data



Plain-English queries powered by Exasol's MCP server.

Banks miss systemic problems until the RBI sees them.
This engine is the direct antidote to escalating mandates.

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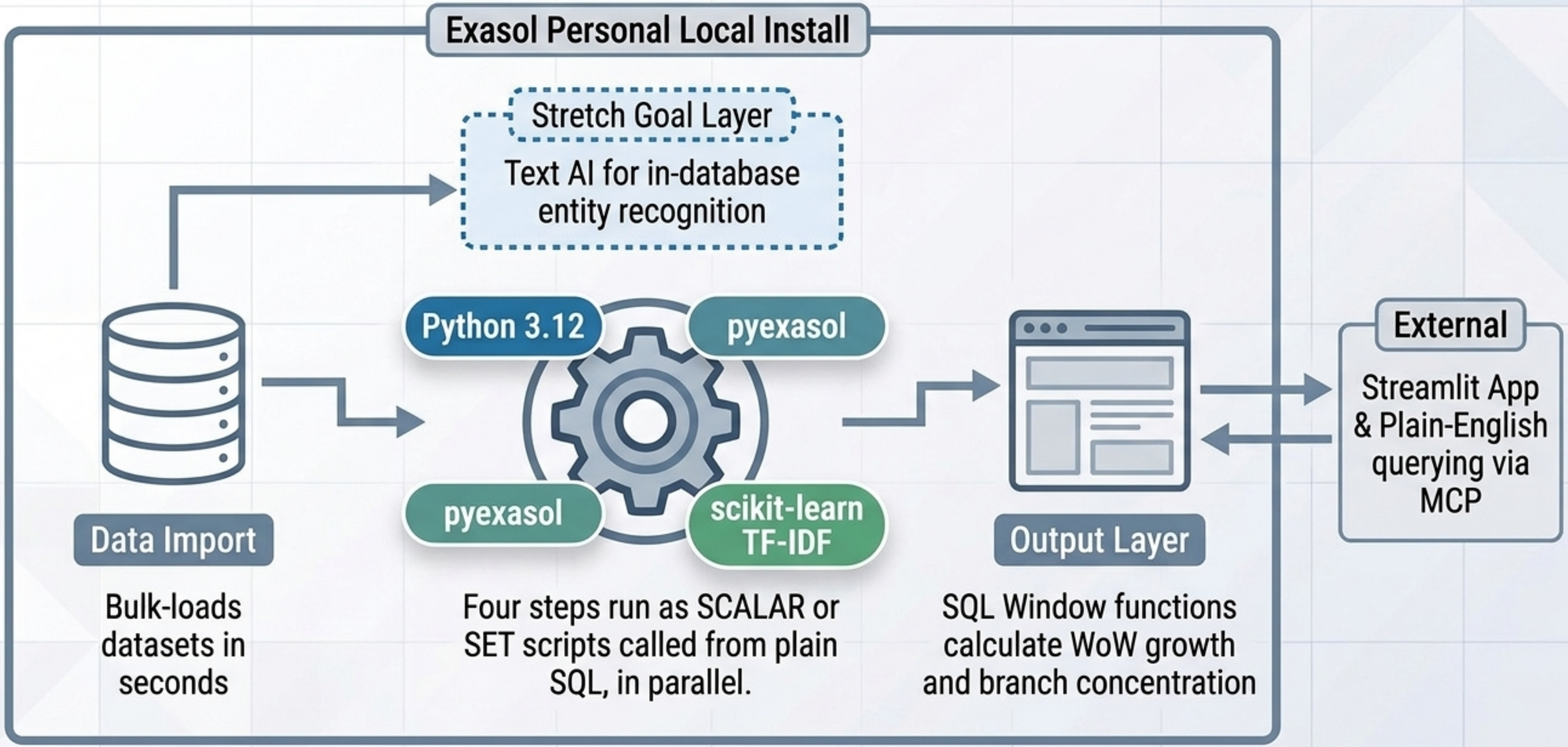
Aligning App Output to RBI Compliance

Dashboard Feature	Engineered Output	RBI Mandate Fulfilled
Root-cause board	Ranked clusters mapped to RBI grounds	Deputy Gov mandate for root-cause analysis
Misclassification watchlist	Tickets closed as ‘query’ with complaint triggers	Governor warning on deliberate misclassification
Escalation gap	Isolates rejected grievances failing to escalate	Governor warning on hidden rejected grievances
Top-five grounds draft	Annual report disclosure table from re-classified NLP data	Fulfills RBI Master List Annual Disclosure requirement
Ask the Data (MCP)	Ad-hoc semantic search on branch/product spikes	Internal operational agility & audit readiness

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Under the Hood: Exasol Architecture Flow

```
> exasol install local
> exasol slc install python3
```



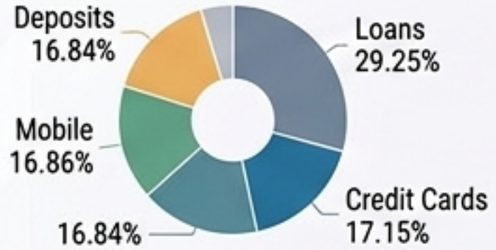
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The Dual-Dataset Validation Strategy

Dataset A: Bharat Co-op Bank



Type:	Synthetic Indian Bank Data
Scale:	60,000 rows (scaled to 2M for speed test)
Design:	Written from templates + LLM paraphrases. 40 branches, 5 channels.



Dataset B: US CFPB Database



Type:	Real Consumer Complaints
Scale:	50,000 genuine human-written narratives
Purpose:	Used strictly to prove the NLP classifier's accuracy on chaotic human writing.
Labels:	CFPB issue fields act as ground truth.

Compliance Check: Rules allow public or synthetic data only. No employer data used. 100% compliant.

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Detecting Planted Patterns in Synthetic Data

If the clustering algorithm works without hand-tuning, these four planted process failures will surface to the top.

1. UPI Reversal Delays



Symptom: Spiking at 8 branches after a core-banking change.

Camouflage: 35% logged as "query".

2. ATM Cash Failures



Symptom: Account debited, no cash dispensed.

Camouflage: Concentrated entirely on one specific ATM vendor.



3. SMS Alert Charges



Symptom: Complaints over minimum-balance and SMS fees after policy change.

Camouflage: Mostly tagged as "request".

4. Loan Interest Resets

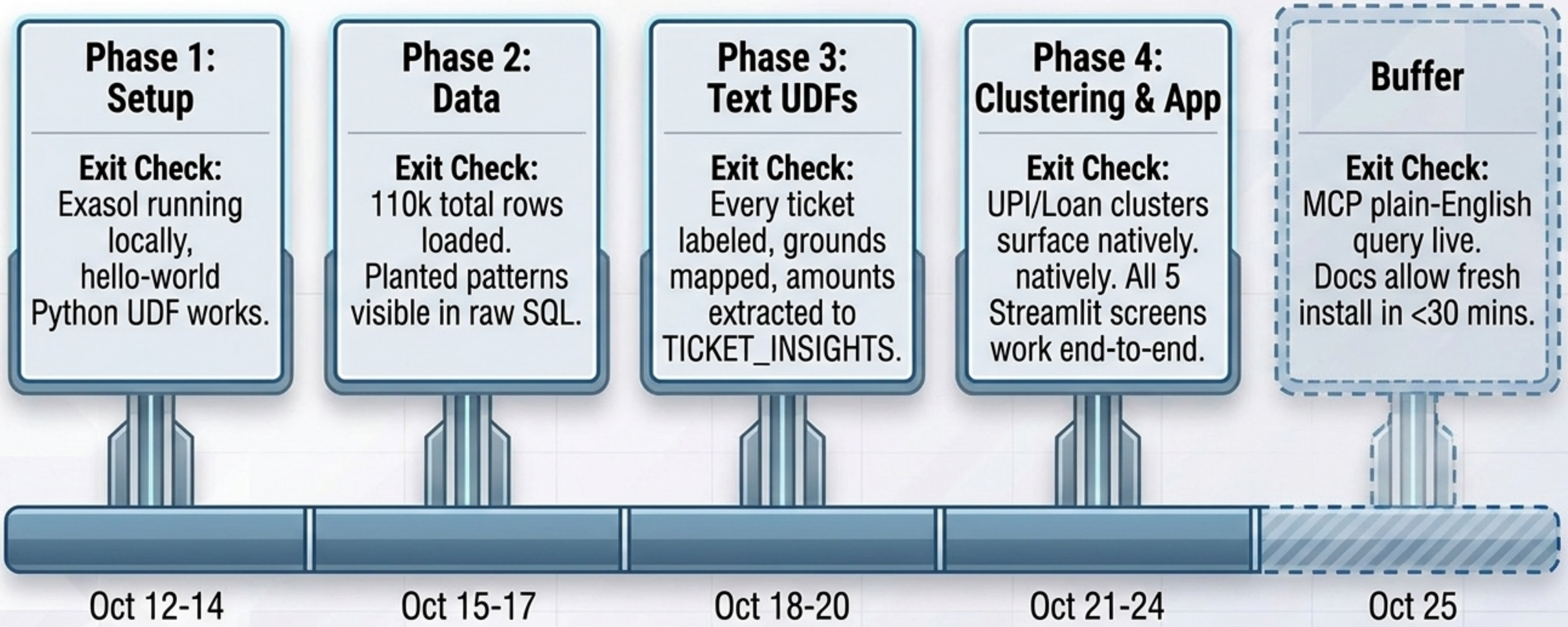


Symptom: Interest reset not communicated.

Validation: The largest cluster, perfectly matching Loans being the RBI's top complaint category.

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Execution Schedule & Exit Checks



Fail-Safe Principle: Each phase ends with a working build. A schedule slip cuts features, not the demo.

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The 3-Minute Demo Storyboard

Agent tags hide a broken process; Exasol finds it directly from the text.

0:00 - 0:25



Establishing the illusion: “RBI says real complaints hide as queries.”

0:25 - 0:50



One SQL call runs UDFs over 60,000 tickets instantly. **Data never leaves the bank.**

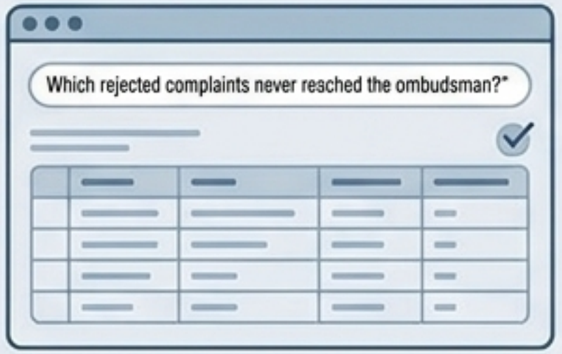
0:50 - 2:05



The **Root-Cause Board** surfaces the broken UPI process (33% of complaints).

A stuck payment tagged as ‘query’ is isolated.

2:05 - 3:00



Live MCP Search: “Which rejected complaints never reached the ombudsman?” Answered live via MCP.

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Technical Defensibility & Fallbacks

Identified Risk ⚠️	Engineered Fallback ✔️
Proprietary Text AI does not install locally on Exasol Personal.	Custom Python TF-IDF UDFs (scikit-learn) with keyword rules. Text AI remains a stretch goal.
Synthetic text is too structured, making clustering clustering accuracy look fake.	Report real-world accuracy on the 50,000 US CFPB dataset as the headline number to prove NLP works on chaotic text.
Local laptop hardware chokes on the 2-million-row scale test during demo.	Pre-provision a Cloud Exasol Personal instance specifically for the speed-test slide.

Finalist Q&A Strategy: Live MCP Juror Testing

We will invite a judge to type their own plain-English query live via MCP, proving the system is dynamic, real-time, and completely unscripted.

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